

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 14: Automated Parameter Calibration

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Overview

A bounded optimization loop updates one parameter block at a time while preserving the established weak-field limits.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$\theta_{r+1} = \theta_r + \Delta\theta_r$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$L_R = \sum_{a=1}^{52} R_a^2$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$score = (L_R, N_{regress}, d_{SGT9})$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Campus Autonomous Research Protocol CARP-7
- Campus Refinement Loop EAC operator guide
- SGT9 issuance revisions 481–512